

Since each hour has 60 minutes therefore contains 12 5-minute intervals, starting from minute 0 until minute 55. Therefore, a set of interval is $H_k = \{0, 5, 10, \dots, 55\}$ where the interval value is v_i for $i \in H_k$. Let define the hourly dispatch price as

$$x_h = \frac{1}{12} \sum_{i \in H_k} v_i \quad (1)$$

where h is the index of the hour, not necessarily be in the 24 hour limit.

However, I have a population of hours in a whole year. I want to estimate that population mean, hence the population parameter is

$$\theta = \frac{1}{N} \sum_{h=1}^N x_h \quad (2)$$

I will use stratified sampling, each stratum is block of hours in a day such as morning peak hours, working hours, evening peak hours, night hours, sleep hours to capture the characteristics of the data set. Therefore, there are 5 strata ($M = 5$). I want to create a sample of arbitrary size n . So my population mean now is,

$$\mu = \frac{1}{N} \sum_{d=1}^M \sum_{h=1}^{N_d} x_{dh} = \sum_{d=1}^M w_d \mu_d \quad (3)$$

where $w_d = N_d/N$, $\mu_d = \frac{1}{N_d} \sum_{h=1}^{N_d} x_{dh}$. The latter summation is the weighted average of strata means μ_d .

Estimation

Let the mean of each stratum d be $\hat{\mu}_d$. In stratified sampling, I choose n_d units from a stratum. Let s be the chosen units such that $s_1 = s_2 = \dots = s_5$ which means that taking exactly same number of units in each stratum. Assume that I have some arbitrary stratum d , s_d units of choice. Therefore, let the number of samples can be taken from the stratum be K then $K = \binom{N_d}{s_d}$, where N_d is the population of the stratum in my case the number of combination is extremely large thus rendering the simple random sampling (SRS) procedure in each stratum impractical.

Instead of choosing all K samples of the stratum, I can choose $\hat{K}$ such that $\hat{K} \ll K$. Now, the expected population parameter is

$$\mathbb{E}(T) = \sum_{k=1}^{\hat{K}} \hat{\theta}_k \pi_k \quad (4)$$

where $\hat{\theta}_k = \bar{x}_{d,k}$, the arithmetic mean of stratum d of sample k . Let $S_{d,k} = \{i_1, i_2, \dots, i_{s_d}\}$, where if $h \neq l$ then $i_h \neq i_l$ and $i_h \in \{1, 2, \dots, N_d\}$. This is a sample k of stratum d . I have that

$$\bar{x}_{d,k} = \frac{1}{s_d} \sum_{i \in S_{d,k}} x_i \quad (5)$$

To calculate bunch of $\bar{x}_{d,k}$'s for $k = 1, 2, \dots, \hat{K}$ and every stratum. In R, I use $\hat{\mu}_h$ via $\bar{x}_{d,k}$ (a biased estimator).

```

1 drawn <- lapply(strata, function(stratum) {
2   stratum[sample(nrow(stratum), s_d, replace=FALSE), ]
3 })
4 # Mean of each sample k of each stratum.
5 mu_d_hat <- sapply(drawn, function(d) mean(d$Mean)) #
```

Since this is SRS, then each sample has the same probability of being selected of $1/K$. From Equation (4) and Equation (5) hence

$$\begin{aligned}\mathbb{E}(T) &= \sum_{k=1}^{\hat{K}} \bar{x}_{d,k} \frac{1}{K} \\ \mathbb{E}(T) &= \sum_{k=1}^{\hat{K}} \sum_{i \in S_{d,k}} x_i \frac{1}{K s_d}\end{aligned}\tag{6}$$

Furthermore, to estimate θ , I need $\hat{\mu}$, which is

$$\hat{\mu} = \bar{x}_k = \sum_{d=1}^M w_d \bar{x}_{d,k}\tag{7}$$

Hence in R it is

```
1 sum(weights*mu_h_hat)
```

Expected Popluation Parameter

Given that each unit has a chosen probability of s_d/N_d . In R, it is not possible to generate the first $\hat{K}$ combinations so I choose randomly by looping $\hat{K}$ times. The expected number of times a unit appears is

$$\mathbb{E}[U_i] = \hat{K} \frac{s_d}{N_d}\tag{8}$$

I will consider two case since the appearance of units is random. Firstly,

$$\begin{aligned}\hat{K} \frac{s_d}{N_d} \sum_{i=1}^{N_d} x_i &\geq \sum_{k=1}^{\hat{K}} \sum_{i \in S_{d,k}} x_i \\ \frac{\hat{K}}{K} \frac{1}{N_d} \sum_{i=1}^{N_d} x_i &\geq \frac{1}{K s_d} \sum_{k=1}^{\hat{K}} \sum_{i \in S_{d,k}} x_i \\ \frac{\hat{K}}{K} \mu_d &\geq \mathbb{E}(T) \\ \frac{\hat{K} - K}{K} \mu_d &\geq \text{bias}\end{aligned}\tag{9}$$

For the other case it is

$$\frac{\hat{K} - K}{K} \mu_d < \text{bias}\tag{10}$$

Through the derivation, somehow the s_d is eliminated but in Equation (9). If I make s_d somehow large then the bias just be $-\mu_d$. Illustrated with the graph below.

$\bar{x}_{d,k}$ remains to be a biased estimator of μ_d . In addition to that, w_d 's are not all equal. Therefore, because of the previous two reasons, $\bar{x}_k$ is a biased estimator of μ . However, by choosing the correct amount of $s_d, \hat{K}$, the bias can be reduced. In R, I use the `replicate(K_hat, {...})` function to repeat the sampling procedure $\hat{K}$ times and calculate the mean of the whole sample each for each sampling. The vector `sample_means` contains the means of the sampling plans, that is, $\{\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_{\hat{K}}\}$ of $\{S_1, S_2, \dots, S_{\hat{K}}\}$ where each has a uniform probability of $\pi_k = 1/K$ for $k = 1, 2, \dots, \hat{K}$.

```
1 sample_means <- replicate(K_hat, {
2   drawn <- lapply(strata, function(stratum) {
```

```
3       stratum[sample(nrow(stratum), s_d, replace=FALSE), ]
4     })
5     # Mean of each sample k of each stratum.
6     mu_h_hat <- sapply(drawn, function(d) mean(d$Mean)) # $
7
8     sum(weights * mu_h_hat)
9  })
```
